MIL-P-16617B(OS)
25 July 1967

Superseding
MIL-P-16617A
25 March 1952

MILITARY SPECIFICATION

# PLASTIC, THERMOSETTING: PULP-FILLED PREFORMS

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy.

## 1. SCOPE

1.1 This specification covers one type of pulp-filled thermosetting plastic preforms for molding.

## 2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications and standards of the issue in effect on date of invitation for bids, form a part of this specification.

### SPECIFICATIONS

FEDERAL

L–P–406 —Plastics, Organic; General Specifications, Test Methods.

LLL–B–631 —Boxes, Fiber, Corrugated (for Domestic Shipment).

NAVY DEPARTMENT

General Specifications for Inspection of Material.

### STANDARD

MILITARY

MIL–STD–129—Marking Shipments.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

## 3. REQUIREMENTS

**3.1 Material.**—The preforms shall consist of an unbleached kraft wood pulp combined with one or more thermosetting phenolic resins.

**3.2 Manufacture.**—The preforms shall be made from a slurry of the pulp-resin mixture, felted by vacuum to a thickness of approximately one quarter inch, after which they shall be dried at a temperature not to exceed 150°F to a moisture content of approximately 5 percent.

**3.3 Physical properties.**—The properties of compression-molded rectilinear panels, made by stacking the required number of preforms of the dimensions of the mold and approximately 0.25 inch in thickness, shall conform to the requirements of table I.

**3.4 Workmanship.**—The material shall be of uniform quality, and the workmanship of the preforms shall be first class in every respect.

## 4. SAMPLING, INSPECTION AND TEST PROCEDURES

**4.1 Sampling.**

4.1.1 *Size of lot.*—For sampling purposes a lot shall consist of one mill run (batch) or the material from one day's run, whichever

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TABLE I.—*Physical properties of molded preforms*

| Property | Requirement |
|---|---|
| Water Absorption: | |
| Weight gain in 24 hrs., max., percent | 1.5 |
| Weight gain in 48 hrs., max., percent | 2.0 |
| Izod impact strength, notched flatwise, min., ft-lb per in of notch. | 2.0 |
| Tensile strength, min., lb/in² | 11000 |
| Flexural strength, min., lb/in² | |
| Edgewise | 15000 |
| Flatwise | 15000 |
| Compressive strength, min., lb/in² | |
| Edgewise | 11000 |
| Flatwise | 34000 |
| Heat distortion temperature, min., °F | 320 |
| Specific gravity | |
| Minimum | 1.35 |
| Maximum | 1.41 |

is more frequent. Material in the lot shall be carefully identified by manufacturer's batch or lot number until final action is taken with respect to acceptance or rejection of the lot.

4.1.2 *Samples.*

4.1.2.1 *Samples for lot acceptance tests.*— The contractor shall furnish to the inspector, from each lot, sufficient rectilinear preform panels approximately 7 by 9 inches by one-quarter inch thick. By stacking the required number of these performs, 6 rectilinear panels 7 by 9 by 0.250 + 0.000–0.050 inches and 7 panels 7 by 9 by 0.500 + 0.000–0.020 inches shall be compression molded and securely packed for shipment to a Government laboratory marked with the following information:

Title, number, and date of this specification.
Samples for lot acceptance.
Lot number.
Manufacturer's designation.
Name of contractor.
Submitted by_____(Name) (date)

4.1.2.2 Samples for lot control tests.—The inspector shall take sample panels for control test as specified in 4.1.2.1, except that only one panel of each thickness shall be required for the control tests specified in 4.3.2.

4.1.2.2.1 When a manufacturer of preforms has furnished two successive lots of preforms which have passed all the tests required for lot acceptance (see 3.3 and 4.3), the inspector may accept successive subsequent lots on water absorption and impact tests only, except for such check tests (see 4.3.3) as he may choose to have run from time to time, until a lot fails to pass either of these tests, or one or more of the check samples fail to meet any of the values shown in table I. When any lot fails to pass on the first performance of any test or set of tests, that lot and all subsequent lots shall be subjected to all of the tests shown in table I until two successive lots thus tested conform to the properties shown in table I.

4.1.2.2.2 When a contractor has consistently furnished preforms conforming to this specification under one or more contracts or orders, control sampling and testing only may be required, providing the contractor certifies that the preforms he proposes to furnish under a new contract or order conform in all respects with the acceptable material which he has previously furnished.

4.2 **Inspection procedures.**— For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

4.3 **Tests.**

4.3.1 *Tests for lot acceptance.*—Tests for lot acceptance shall be made on specimens machined from the sample panels submitted for tests, in accordance with the following test methods of specification L–P–406:



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| | *Method No.* |
|---|---|
| Water absorption | 7031 |
| Izod impact strength | 1071 |
| Tensile strength | 1011 |
| Flexural strength | 1031 |
| Compressive strength | 1021 |
| Heat distortion temperature | 2011 |
| Specific gravity | 5011 |

Unless otherwise specified by the procuring activity, five specimens for each of the above tests shall be required to conform to the requirements shown in table I.

4.3.2 *Tests for lot control.*—Control tests shall be made on specimen machined from panels representative of the lot to the satisfaction of the inspector, at the place of manufacture, in accordance with test methods of Specification L–P–406, for the following properties:

¼-inch panel tested for water absorption, method 7031.

½-inch panel tested for Izod impact strength, method 1071.

Unless otherwise specified by the procuring activity, three specimens for each of the above tests shall be required to conform to the requirements for these properties shown in table I.

4.3.3 *Check tests.* — The inspector shall from time to time forward test panels to a Government laboratory for the purpose of checking for any or all of the properties shown in table I. (See 4.1.2.2)

4.3.4 *Action in case of rejection.*—A lot that has been rejected under the provisions of this specification may be resubmitted for inspection and testing, provided the contractor, after being informed of the reasons for rejection, has suitably reworked the entire lot and has corrected the deficiencies noted. When resubmitted for inspection and testing the lot shall be sampled in accordance with the procedure outlined in 4.1.2.1 and two successive lots thus sampled, and tested in accordance with 4.3.1, shall be required to conform to the properties shown in table I before permission is granted to return to lot control sampling.

## 5. PREPARATION FOR DELIVERY

**5.1 Packaging.**—Unless otherwise specified by the procuring activity, preforms covered by this specification shall not be individually packaged prior to shipment.

**5.2 Packing (domestic shipment).** — Unless otherwise specified, preforms shall be packed in 275-lb. min. Mullen test corrugated-fiber boxes complying with Specification LLL–B–631. Preforms shall be separated by means of dividers adequate to prevent damage in handling or shipment. The gross weight of the carton shall not exceed 70 pounds.

**5.3 Marking.**—In addition to any special marking required by the contract or order, each container shall be marked in accordance with MIL–STD–129.

## 6. NOTES

**6.1 Ordering data.** — Procurement documents should specify the following:

Title, number and date of this specification.

Dimensions of the preforms.

**6.2** When this specification is used for the procurement of preforms for the manufacture of plastic cartridge-case plugs, NAVORD OS 3824, the preforms shall be circular discs approximately one inch less in diameter than the mold cavity.

**6.3** The following formulations have been found to meet the requirements of this specification and are included for information only:

Unbleached kraft (wood pulp) 50% by weight

Durez No. 13007 (or equal) 50% by weight



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Unbleached kraft (wood pulp)
50% by weight
Durez No. 10650 (or equal)
25% by weight
Durez No. 11078 (or equal)
25% by weight

**Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.**

**Custodians:**

Navy—OS



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PLATE NO. 15419 (BACK)

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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

**INSTRUCTIONS**

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

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